Meta-Dependence in Conditional Independence Testing

Bijan H. S. Mazaheri [1,2] Jiaqi Zhang [2,3] Caroline Uhler [2,3]

[1]Dartmouth Engineering [2]Eric & Wendy Schmidt Center, Broad Institute of MIT and Harvard [3]LIDS, MIT

POSTER PAPER CODE

The problem. Causal discovery, feature screening, and invariance testing chain together *many* conditional independence (CI) tests, and an algorithm is wrong when *at least one* of them is wrong. So co-occurring failures are cheaper than independent ones — but Bonferroni-style corrections *assume* independence, and the graphoid axioms (Pearl and Paz, 1986) constrain CIs only *qualitatively, at the level of the graph*.

This paper. We define **CIMD**, a measure of "meta-dependence" built from *moment projections*: no causal graph, no raw data, and for Gaussians a closed form in the **covariance matrix alone**. CIMD is signed, is *not* fixed by the graph, tracks sub-sampling behaviour, and yields a redundancy-aware multiplicity correction for PC.

1 Conditional independence tests do not fail independently

① Sub-sample the data $1000\times$

i	A	B	C
1	-0.4	0.2	1.1
2	1.2	0.9	0.3
3	0.5	-1.3	-0.8
4	-1.1	-0.6	0.4
5	0.8	1.5	1.2
6	0.1	-0.2	-1.4
7	-0.7	0.4	0.9
8	1.6	1.1	-0.2
9	-0.3	-0.9	0.6
10	0.9	0.7	1.3
11	-1.4	0.3	-0.5
12	0.2	-1.1	0.8

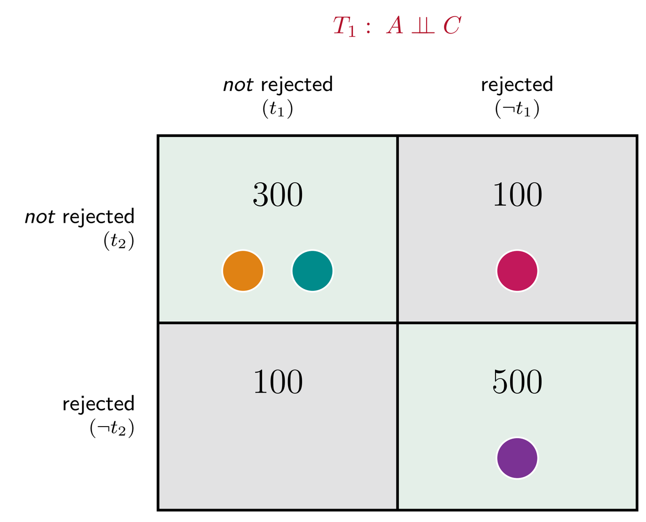
sub-sample 1 2 3 4

one colour = one random sub-sample of size $n \ll N$

② run *both* CI tests on *every* sub-sample

record which *pair* of outcomes occurred

③ Tally the pairs of outcomes



dots: the four sub-samples at left
numbers: all 1000 sub-samples

④ Read off the dependence

$$\text{FS-CID} := \hat{P}(t_1, t_2) - \hat{P}(t_1)\,\hat{P}(t_2)$$
$$0.30 - (0.40)(0.40) = +\mathbf{0.14}$$

Independent tests would leave only 160 sub-samples in the top-left cell; we observe 300.

The two tests fail *together*. A negative FS-CID is also possible: the tests can instead fail on *disjoint* sub-samples.

Sub-sample a dataset repeatedly and each CI test becomes a *random variable*. "Finite-Sample CI Dependence" (FS-CID) asks how the outcomes of two such tests co-occur — and they do, in both directions.

An algorithm that chains CI tests is wrong when *at least one* test is wrong, so *independent* failures are costlier than co-occurring ones. Bonferroni divides α by the number of tests: it *assumes* independent errors, and so over-corrects redundant ones.

Graphoid axioms (Pearl and Paz, 1986) and Verma constraints (Verma and Pearl, 1990) do capture how CIs constrain one another, but only *qualitatively, at the level of the graph*.

FS-CID is a ground truth rather than a tool: each value costs a thousand re-runs of *both* tests, repeated for every pair of tests one cares about. And it explains nothing — it cannot say *which* distributions produce which sign, or why.

The rest of this poster replaces it with a quantity that is predicted by geometry (2), computed once from a covariance matrix (3), and spent as a multiplicity budget (4).

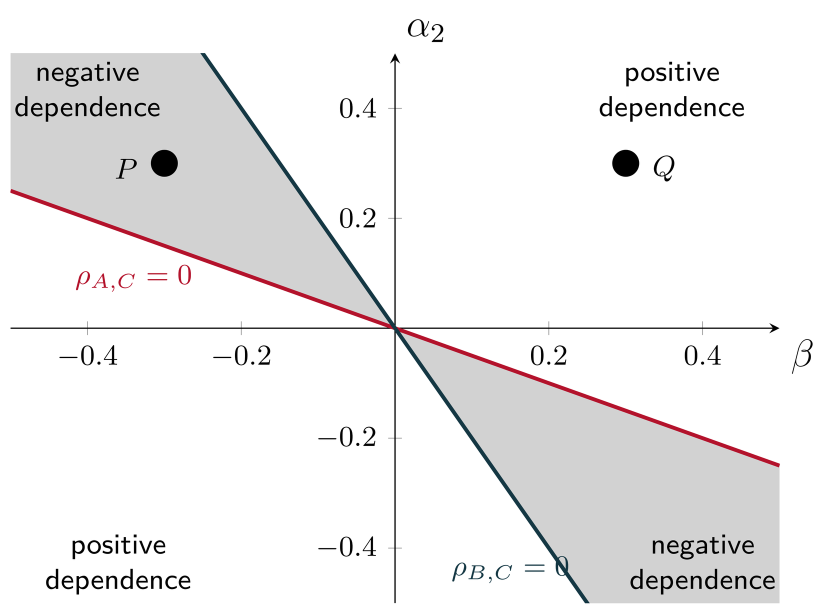
2 It comes from the geometry of CI manifolds

Standardized linear–Gaussian model on the fully connected DAG $A \to B \to C$, $A \to C$:

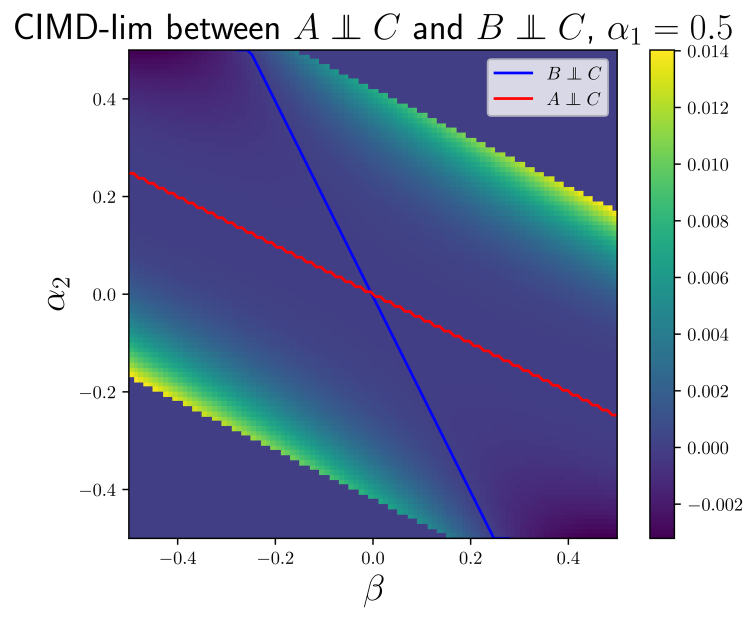
$$A = N_A,$$
$$B = \alpha_1 A + N_B,$$
$$C = \alpha_2 A + \beta B + N_C,$$

$$\rho_{A,B} = \alpha_1,$$
$$\rho_{A,C} = \alpha_2 + \alpha_1\beta,$$
$$\rho_{B,C} = \alpha_1\alpha_2 + \beta.$$

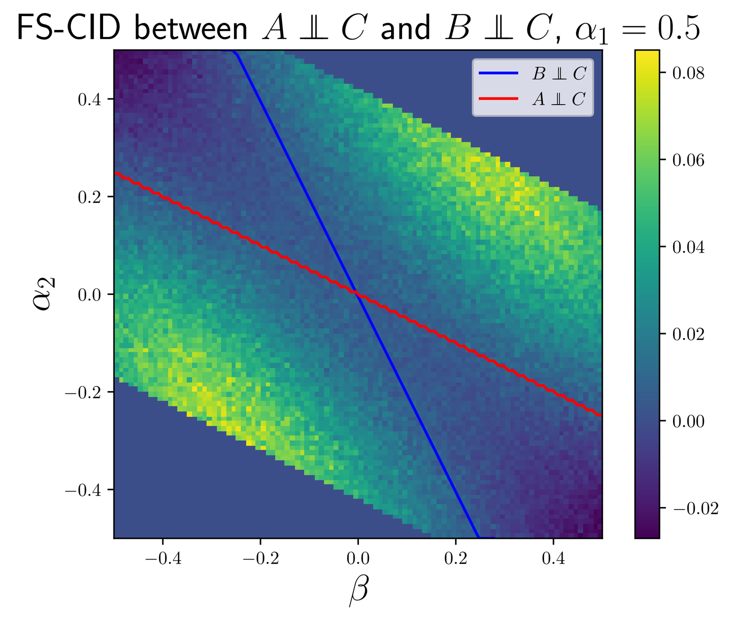
Each CI property is a **manifold** $\{\rho = 0\}$ — here a line. A perturbation moves P toward both at once (*positive* dependence) or toward one and away from the other (*negative*).



(a) parameter space, $\alpha_1 = 0.5$



(b) CIMD: covariance matrix only



(c) FS-CID: 1000 sub-samples of 20 points

The geometry's prediction **(a)** is recovered by our closed-form measure **(b)** and by the finite-sample ground truth **(c)** — though **(b)** never touches the data. Real data agrees too (Apple Watch & Fitbit (Fuller, 2020), California Housing, Auto MPG). $P\,(\beta < 0)$ and $Q\,(\beta > 0)$ obey the *same DAG* yet have *opposite-sign* meta-dependence: not a graphical phenomenon.

3 CIMD: meta-dependence from a covariance matrix

Project into independence

The **M-projection** of P onto a set $\mathcal{P}$ is its KL-closest member (Nielsen, 2018), $\Pi_{\mathcal{P}}(P) := \arg\min_{Q\in\mathcal{P}} D(P \,\|\, Q)$; onto the distributions Markov to a graph $\mathcal{G}$ it is just $\mathcal{G}$'s factorization built from P's own conditionals (Koller and Friedman, 2009). For a *single* CI, project onto the **auxiliary fork** $A \leftarrow \mathbf{C} \to B$:

$$\left[\Pi_{A \perp\!\!\!\perp B \mid \mathbf{C}}(P)\right](a, b, \mathbf{x}) = \mathbb{E}_{\mathbf{C}}\left[P(a \mid \mathbf{C})\,P(b \mid \mathbf{C})\,P(\mathbf{x} \mid a, b, \mathbf{C})\right].$$

The true graph never enters — the auxiliary graph is a placeholder chosen only to induce the target independence.

CIMD-lim

When either mutual information is large (> 0.1 nats) the test essentially never fails to reject, so no finite-sample dependence is resolvable there. **CIMD-lim** gates CIMD to zero in that regime, keeping only the near-violation region where CI tests actually flip. It is what panels (b) and the real-data figures plot.

Measure the shrinkage

Definition (CIMD)

For (T_1) $A \perp\!\!\!\perp B \mid \mathbf{C}$ and (T_2) $A' \perp\!\!\!\perp B' \mid \mathbf{C}'$,

$$\mathrm{CIMD}(T_1, T_2, P) := I_P(A : B \mid \mathbf{C}) - I_{\Pi_{T_2}(P)}(A : B \mid \mathbf{C}).$$

T_1 fixes *which* mutual information we watch; T_2 fixes *which* projection comes first.

A projection is the *minimal* perturbation of P onto a CI property; CIMD asks whether it also moved P toward a second one.

- > 0 the manifolds point the same way, so T_1 adds little once T_2 is known;
- $= 0$ the two tests are "orthogonal";
- < 0 the manifolds lie on opposite sides of P.

Panel (a) above shows both signs.

Asymmetric: in $A \to B \to C \to D$, perturbing toward $B \perp\!\!\!\perp C$ forces $A \perp\!\!\!\perp D$ but not conversely, so $\mathrm{CIMD}(T_1, T_2, P)$ asks: is T_1 redundant once T_2 is known?

A Gaussian closed form

Let Σ be the covariance of P over $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{X}$. The projected $\Sigma^{\perp\!\!\!\perp}$ agrees with Σ except that the $\mathbf{A}, \mathbf{B}$ conditional covariance given $\mathbf{C}$ is zeroed,

$$\Sigma^{\perp\!\!\!\perp}_{\mathbf{A}\cup\mathbf{B}} = \Sigma_{\mathbf{A}\cup\mathbf{B},\mathbf{C}}\Sigma^{-1}_{\mathbf{C},\mathbf{C}}\Sigma_{\mathbf{C},\mathbf{A}\cup\mathbf{B}},$$

and that change is propagated through $P(\mathbf{X} \mid \mathbf{A}, \mathbf{B}, \mathbf{C})$. With $\Phi, \Phi^{\perp\!\!\!\perp}$ the conditional covariance of $\mathbf{Z} = \mathbf{A} \cup \mathbf{B}$ given $\mathbf{C}$ before and after,

Closed form

$$\mathrm{CIMD} = \tfrac{1}{2}\log\frac{\det(\Phi_{\mathbf{A},\mathbf{A}})\det(\Phi_{\mathbf{B},\mathbf{B}})}{\det(\Phi)} - \tfrac{1}{2}\log\frac{\det(\Phi^{\perp\!\!\!\perp}_{\mathbf{A},\mathbf{A}})\det(\Phi^{\perp\!\!\!\perp}_{\mathbf{B},\mathbf{B}})}{\det(\Phi^{\perp\!\!\!\perp})}$$

Summary statistics suffice: no raw sample is needed, which matters for large or privacy-restricted data where Σ is available but the rows are not.

4 Use case: a redundancy-aware multiplicity correction for PC

PC deletes an edge $\{A, B\}$ if *any* conditioning set separates them, so every edge carries a family of tests: $A \perp\!\!\!\perp B$, $A \perp\!\!\!\perp B \mid C$, $A \perp\!\!\!\perp B \mid D$, $A \perp\!\!\!\perp B \mid \{C, D\}$, … Bonferroni divides α by all n_{edge} of them, as though each were an independent look — but overlapping conditioning sets probe overlapping dependencies. Averaging the normalised, gated CIMD over pairs of conditioning sets gives a mean redundancy $\bar{r} \in [0, 1]$, so we spend only the *effective* number of tests:

Redundancy-relaxed correction

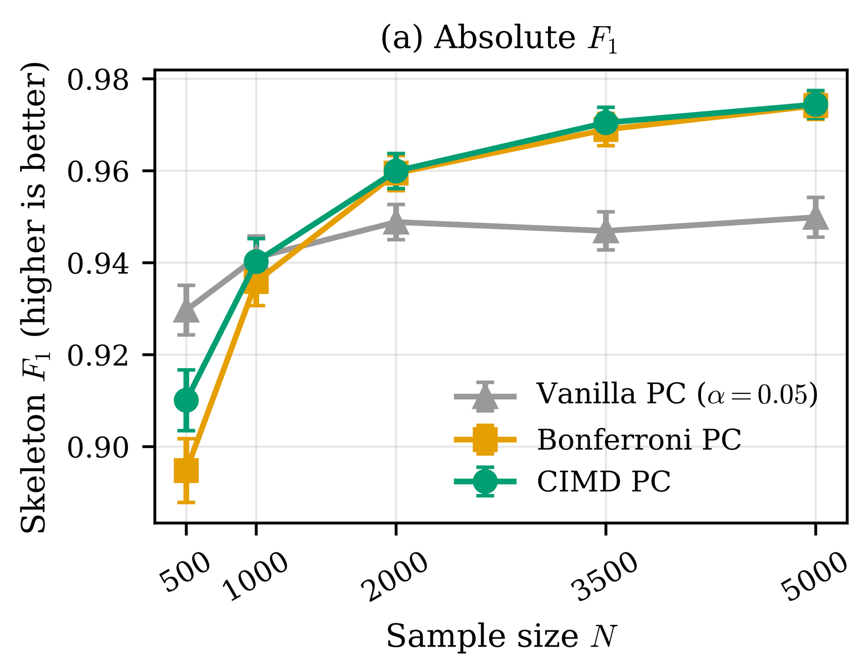
$$n_{\text{eff}} = n_{\text{edge}}\,(1 - \bar{r}), \qquad \text{measured } \bar{r} \approx 0.6\text{–}0.7$$

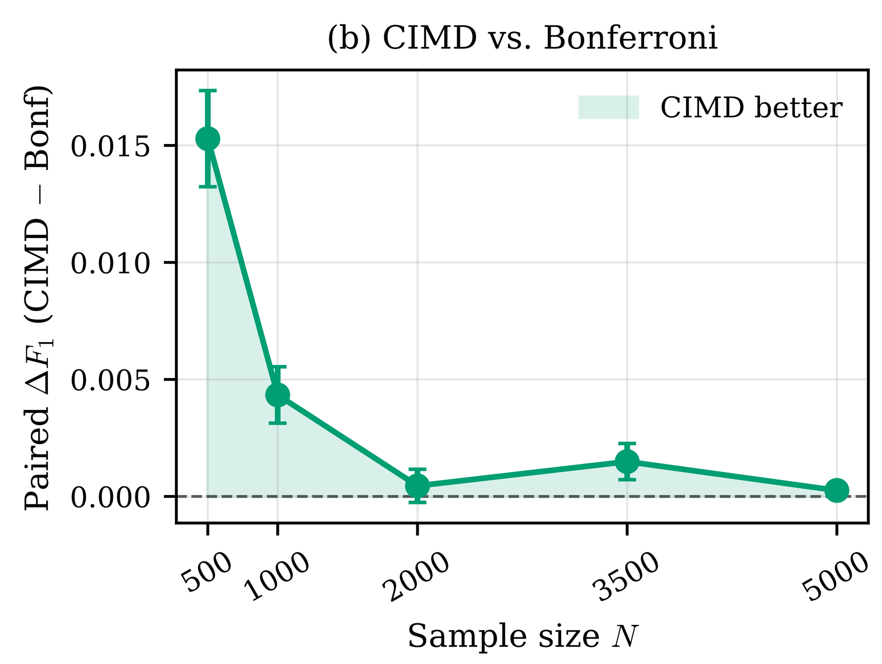
so only 30–40% of an edge's tests count as a fresh look; $\bar{r} = 0$ is exactly Bonferroni.

CIMD-PC never loses to Bonferroni-PC, and gains where the two differ most — the small-sample regime: $\Delta F_1 = +0.015 \pm 0.002$ and 0.47 ± 0.06 fewer skeleton errors at $N = 500$, converging for $N \geq 2000$.

The modest claim: *when a correction is applied anyway*, accounting for redundancy beats assuming independence — not exact finite-sample error control, which is infeasible even for one CI test (Shah and Peters, 2020).

Nothing here is specific to PC: the same redundancy score could tune any procedure that spends a multiplicity budget over a family of overlapping CI tests.





Sparse linear–Gaussian DAGs ($p = 18$, conditioning sets up to size 3, Fisher-Z at $\alpha = 0.05$, 100 random graphs per point, ± 1 SEM). Because normalised CIMD is a function of Σ, the correction costs no extra passes over the data.

Open questions

(i) A *formal* CIMD–FS-CID link needs control of CI-manifold curvature near P, whose geometry is so far described mainly through its singularities (Uhler et al., 2013). **(ii)** Non-Gaussian computation: kernel MI estimators, or normalizing-flow projections onto CI manifolds. **(iii)** Distribution-aware losses for validating causal discovery: count *co-occurring* edge errors rather than raw structural differences.

Acknowledgements

We thank Murat Kocaoglu, Abhinav Kumar, Beshr Islam Bouli, and the anonymous reviewers. B.M.: DARPA ARC COMPASS (HR001-25-3-0212). J.Z., C.U.: Eric & Wendy Schmidt Center, NCCIH/NIH (1DP2AT012345), ONR (N00014-24-1-2687), DOE-ASCR (DE-SC0023187).

References

Mazaheri, Zhang & Uhler. *Meta-Dependence in Conditional Independence Testing.* Proc. 42nd Conf. on Uncertainty in Artificial Intelligence, PMLR 337:4427–4440, 2026.

Chandler Squires. *causaldag: creation, manipulation, and learning of causal models*, 2018. URL https://github.com/uhlerlab/causaldag.

D. Fuller. Replication Data for: Using machine learning methods to predict physical activity types with Apple Watch and Fitbit data using indirect calorimetry as the criterion., 2020. URL https://doi.org/10.7910/DVN/ZS2Z2J.

D. Koller and N. Friedman. *Probabilistic graphical models: principles and techniques*. MIT press, 2009.

F. Nielsen. What is an information projection. *Notices of the AMS*, 65(3):321–324, 2018.

J. Pearl and A. Paz. GRAPHOIDS: Graph-Based Logic for Reasoning about Relevance Relations OrWhen Would x Tell You More about y If You Already Know z? *Probabilistic and Causal Inference*, 1986. URL https://api.semanticscholar.org/CorpusID:262089256.

R. D. Shah and J. Peters. The hardness of conditional independence testing and the generalised covariance measure. *The Annals of Statistics*, 48(3), June 2020. ISSN 0090-5364. doi: 10.1214/19-AOS1857.

P. Spirtes, C. N. Glymour, and R. Scheines. *Causation, prediction, and search*. MIT press, 2000.

C. Uhler, G. Raskutti, P. Bühlmann, and B. Yu. Geometry of the faithfulness assumption in causal inference. *The Annals of Statistics*, pages 436–463, 2013.

T. Verma and J. Pearl. Causal networks: Semantics and expressiveness. In *Machine intelligence and pattern recognition*, volume 9, pages 69–76. Elsevier, 1990.